Assignment - 3

Q1. Explain Machine learning and how it is beneficial in today's era?

Ans. Machine learning is a branch of Artificial Intelligence that enables systems to learn from data and improve their performance without being explicitly programmed.

Instead of following fixed instructions, ML models identify patterns in data and make predictions or decisions automatically.

How it works -

- Data is Collected
- Model is trained on data
- Patterns are learned
- Predictions are made on new data.

→ Benefits in Today's Era

1. Automation of tasks
ML reduces human effort by automating repetitive tasks like email filtering and customer support.

Teacher's Signature: . . . . .

Page No.:
Date: / /

2. Better decision Making
Businesses use ML for data-driven decisions (e.g Demand forecasting).

3. Personalization
Platforms like netflix and amazon use ML for recommendations.

4. Healthcare Advancements.
ML helps in disease prediction, diagnosis and drug discovery.

5. Fraud Detection.
Banks use ML to detect unusual transaction patterns.

Q2. Explain different Supervised and unsupervised ML algorithms.

Ans. A. Supervised Learning Algorithms

1. Linear Regression

- Used for predicting continous values.

- Examples: Predicting house prices

2. Logistic Regression

- Used for classification problems.
- Examples: Spam or not spam.

3. Decision Trees
   - Tree-like structure of decisions
   - Easy to interpret

4. Support Vector Machines (SVM)
   - Finds optimal boundary b/w classes
   - works well with high-dimentional data.

5. K-Nearest Neighbors (K-NN)
   - classifies based on closes data points.

B. Unsupervised Learning Algorithms

1. K-Means Clustering
   - Group data into clusters
   - Example: Customer segmentation.

2. Hierarchical Clustering

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- Builds clusters in a tree-like structure.

3. Principal Component Analysis (PCA)

- Reduces dimentionality of data.

4. Association Rule learning

- Finds relationships b/w variables
- Example : Market basket analysis.

Q3. Differentiate between Supervised and Unsupervised ML Algorithms.

Ans.

Supervised Learning	Unsupervised Learning
1. Type of data is labeled data.	1. Type of data is Unlabelled data.
2. The goal is to predict output.	2. The goal is to find patterns.
3. Examples are Classification, Regression.	3. Examples are Clustering, association.
4. Complexity : It is easier to evaluate.	4. Complexity is harder to interpret.
5. Known output	5. Unknown Patterns.

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Page No.:
Date: \ \

Q4. Explains Feature Selection algorithms - Filters, Wrappers.

Ans. Feature Selection is the process of selecting important features (variables) from a dataset to improve model performance.

A. Filter Methods

- Select Features based on statistical properties.
- Independent of ML models.

Examples

- Correlation coefficient
- Chi - square Test
- Information gain

Advantages

- Fast and Computationally efficient.
- Works well with large datasets.

Disadvantages

- Ignores interaction between features.

Page No.:
Date: / /

B. Wrapper Methods

Uses ML Model performance to select features.

Examples
- Forward Selection
- Backward Elimination
- Recursive Feature Elimination (RFE)

Advantages:
- More accurate than filters
- Considers feature interactions.

Disadvantages
- Computationally expensive
- Risk of overfitting.

Q5. Explain Decision Trees; Random Forests Algorithms.

Ans. A. Decision Tree

A decision Tree is a supervised learning algorithm used for classification

and regression.

Structure -

- Root Node → starting point
- Decision Nodes → Conditions
- Leaf Nodes → final output

Working -

1. Select best Feature
2. Split data based on condition
3. Repeat recursively
4. Stop when criterial met

Advantages -

- Easy to understand
- No need for data normalization.

Disadvantages -

- Prone to overfitting
- Sensitive to data changes.

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B. Random Forests

Random Forests is an ensembling learning method that combines multiple decision trees.

Working

1. Create mutiple random datasets

2. Train multiple decision trees

3. Combine outputs (majority voting or averaging)

Advantages -

- Reduces overfitting
- High accuracy
- Handles large datasets.

Disadvantages

- More Complex
- Slower than single trees.

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